

Name:	ID
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Answer the following questions:

1. Casting pattern is defined as :

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2. Casting mold is defined as:

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3. The advantages of continuous casting are:

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4. The types of continuous casting machines are :

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5. List five advantages of metal casting:

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6. It is required to sand cast the shown product from gray cast iron, knowing that the product will be machined after casting on the surfaces with (▼) mark only. Find: 1. Suitable flask dimensions.

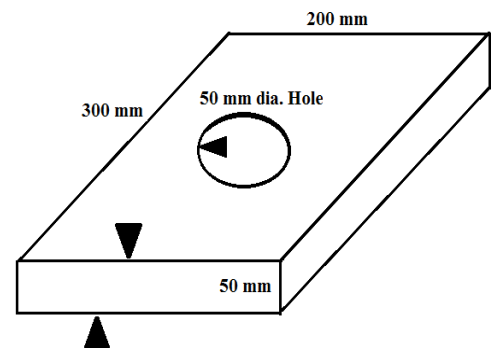
2. Core dimensions.

3. Casting pattern dimensions.

4. The forces acting on core.

5. The forces acting on cope.

6. The forces acting on mold. Take: Shrinkage allowance = 3 mm/m, Machining allowance = 2 mm



A series of horizontal dotted lines for writing.

تأكد من تسجيل الإسم ورقم الفصل باللغة العربية بخط واضح (استخدم ظهر ورقة الإمتحان للحل عند الحاجة لذلك)

Name:

Sec:

[1] Calculate [F] the force needed to break the Joint

[5 Marks]

Weld Size = 10 mm & SMAW Electrode E7018, Note that: [1 Ksi = 6.89 MPa] & [$\tau = \sim 0.6 \sigma$]

Weld Throat $t_w =$
$$[mm]$$

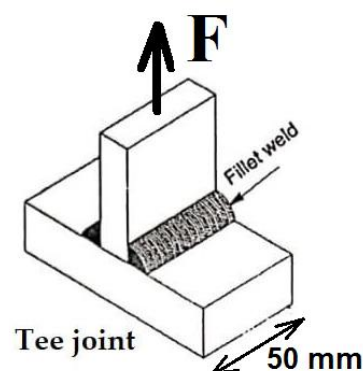
Weld Area Subjected to Shear: $A_S =$

$$[\text{mm}^2]$$
Joint Shear Strength: $\tau =$

[MPa]

Force needed to break the Joint: $F =$

[N]



[2] Explain) the Elements of Welding Symbol

[5 Marks]

